

10. (2.5 points)

Given the following information as of December 31, 2017:

Accident Year	Incremental Paid Loss (\$000) as of (months)	
	12	24
2013	600	1,300
2014	650	1,400
2015	550	1,400
2016	700	1,450
2017	700	

- Exposures are stable over the years.
- There are no calendar or accident date trends in the data.
- A simple linear regression $y = a + bx$ of the incremental loss at age 24 over the cumulative loss at age 12 was performed. The resulting regression coefficients are:
 - $a = 1,075$
 - $b = 0.5$
- For a simple linear regression $y = a + bx$, the standard error for the slope (b) and intercept (a) are calculated as:

$$s_b = \sqrt{\frac{\sum_i \epsilon_i^2}{(n-2) \sum_i (x_i - \bar{x})^2}}$$

$$s_a = s_b \sqrt{\frac{1}{n} \sum_i x_i^2}$$

where ϵ_i are the error terms.

a. (2 points)

Test for significance of the 12-24 loss development factor and state the chain ladder model assumption that this test is used to validate.

b. (0.5 point)

Discuss how the estimated future loss emergence for a year with abnormally high losses to date would differ under the chain ladder versus additive model.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 10

TOTAL POINT VALUE: 2.5

LEARNING OBJECTIVE(S): A3

SAMPLE ANSWERS

Part a: 2 points

Sample 1

x	y	$\hat{y}$	$ \epsilon $
600	1,300	1,375	75
650	1,400	1,400	0
550	1,400	1,350	50
700	1,450	1,425	25

$$\sum (x_i - \bar{x})^2 = 12,500$$

$$s_b = [\sum (\epsilon_i^2 / 25,000)]^{0.5} = [(1 / 25,000) \times (75^2 + 50^2 + 25^2)]^{0.5} = 0.592$$

$$\sum x_i^2 = 1,575,000$$

$$s_a = s_b \times (1 / 4 \times \sum x_i^2)^{0.5} = 0.592 \times (393,750)^{0.5} = 372$$

$s_b > b / 2 \rightarrow$ insignificant

$s_a < a / 2 \rightarrow$ significant

Assumption: the variance of expected future losses in the next development period are proportional to losses to date.

Sample 2

AY	x	Predicted	Actual	$\epsilon = A - \text{Predicted}$
13	600	1,375	1,300	-75
14	650	1,400	1,400	0
15	550	1,350	1,400	50
16	700	1,425	1,450	25

$$Y = a + bx$$

$$a = 1,075$$

$$b = 0.5$$

$$\bar{x} = 625$$

$$\sum x_i^2 = 1,575,000$$

$$\sum (x_i - \bar{x})^2 = 12,500$$

$$s_b = \sqrt{\frac{1}{2 \times (12,500)}} \times (75^2 + 50^2 + 25^2) = 0.592$$

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

$$S_a = 0.592 \times \sqrt{\frac{1,575,000}{4}} = 371.5$$

$a = 1,075 > 2 \times S_a = 2 \times 371.5 = 743$; a is significant

$b = 0.5 < 2 \times S_b = 2 \times 0.592$; b is not significant

Assumption: to validate if incremental losses in next period is proportional to the losses reported to date.

Sample 3

Chain ladder assumption tested: expected loss of next development period is proportional to loss-to-date.

$$S_b = \sqrt{\sum \frac{\varepsilon_i^2}{(n-2)\sum(x_i - \bar{x})^2}}$$

$$\bar{x} = 625$$

$$n = 4$$

$$\sum(x_i - \bar{x})^2 = 12,500$$

AY	$\hat{Y}_{24}$	ε
2013	1,375	-75
2014	1,400	0
2015	1,350	50
2016	1,425	25

$$S_b = \sqrt{\frac{8,750}{(2) \times 12,500}} = 0.592$$

$0 > b - 1.96 \times 0.592$: insignificant

Part b: 0.5 point

Sample responses:

- Abnormally high losses to date would make chain ladder's estimate abnormally high since it is very reflective to losses to date. Additive model would not get impacted as much as C-L since the later years' future development for each period are independent and are not impacted by loss to date.
- C-L will assume future loss proportional to loss to date, so abnormally high loss to date will result in abnormally high loss for the next year. Additive model loss development will not be affected, as it is not linked to loss to date.
- Additive model would not be influenced by the reported to date at an unusually high year. Chain Ladder model would have future losses in proportion to the unusually high

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

losses to date, so the estimated future losses would be high.

- Chain ladder applies a multiplicative factor to losses to date, whereas the additive model adds a flat amount to losses known to date. The CL method would imply that since losses to date have been high, future losses will also be high, so the resulting estimate will be leveraged up. The additive model will have the same estimate of future losses regardless of level of losses reported to date.

EXAMINER'S REPORT

Candidates were expected to know the key assumptions of the chain ladder and additive reserving models, how to test these underlying assumptions, and be able to determine whether a particular model is or is not appropriate for the given situation.

Part a

Candidates were expected to calculate s_b , including all of the required components such as ϵ^2 and $(x_i - \bar{x})^2$. They were expected to recognize that because the slope was insignificant (since it was about the same as the standard error) that the LDF was statistically insignificant. Additionally, they were expected to know the underlying assumption of the chain ladder method.

Common mistakes included:

- Not stating the significance conclusion.
- Not stating the assumption.
- Not taking the square root of s_b^2 .

Part b

Candidates were expected to know that the future loss emergence for a year with abnormally high losses would be higher under a chain ladder methodology compared to an additive methodology. They were also expected to be able to explain why this was true.

Common mistakes included:

- Not providing any explanation as to why the conclusion was true.